

Process FMEA & Quality Risk Assessment

Illustrative pharmaceutical FMEA showing how failure modes and controls drive validation strategy.

Document ID	PQ-DEMO-FMEA-001
Version	01
Discipline	Quality risk management
Deliverable type	FMEA / risk assessment
Status	Fictional demonstration sample
Prepared for	Public ProcessQuill portfolio

This document is intentionally fictional. It demonstrates document architecture, technical reasoning and presentation style only. It does not contain employer, client or proprietary data and must not be used as an approved GMP instruction, protocol or validation record.

1. Purpose and scope

This sample FMEA demonstrates how process risk should drive validation depth rather than exist as a separate scoring exercise. It covers selected failure modes for a fictional sterile-filtration and aseptic-transfer process.

2. Scoring model

- Severity (S): 1 = negligible impact, 5 = potential critical product-quality or patient impact.
- Occurrence (O): 1 = remote, 5 = frequent / expected without additional control.
- Detection (D): 1 = almost certain detection before impact, 5 = low probability of timely detection.
- Illustrative RPN = S x O x D. The example uses qualitative judgment alongside the numeric score.

Numeric thresholds are intentionally illustrative. A real QRM procedure must define the approved scoring and escalation method.

3. Example FMEA table

Failure mode	Potential effect	S	O	D	RPN	Existing controls	Validation / action
Incorrect filter installed	Loss of validated retention performance	5	2	2	20	Filter ID verification; approved BOM; pre-use integrity	Maintain; verify traceability in batch record
Filter damaged during setup	Potential loss of sterility assurance	5	2	3	30	Visual check; pre-use integrity; controlled assembly	Challenge setup and integrity sequence during validation
Differential pressure exceeds limit	Filter stress / reduced process control	4	2	2	16	Pressure monitoring; alarm / stop limit	Verify alarm and defined operating envelope
Excessive bulk hold before filtration	Bioburden growth before sterile filtration	4	3	3	36	Approved hold time; temperature control	Dedicated hold-time evidence / monitoring
Aseptic connection compromised	Post-filter contamination	5	2	4	40	Qualified connection method; Grade A protection; operator qualification	Include intervention challenge in APS / training

4. Risk-based validation decisions

- High-severity risks should influence validation even where occurrence is low.
- Controls that rely on operator technique should be challenged through training, observation or APS rather than accepted only on procedure text.
- Monitoring limits and alarms should be verified where they are credited as preventive or detective controls.
- Residual risk should be documented after actions and linked to routine control strategy.

5. Conclusion

A useful FMEA creates a direct line from failure mode to control to validation evidence. This sample therefore treats risk ranking as an input to test strategy, not as an isolated table.